

# Aakanksha Khandwaha

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## Education

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### Doctor of Philosophy (PhD), Computer Science

Sep 2026 - present

University of Waterloo

- *Advisor:* Dr. Edith Law
- Researching human-AI interaction and human-centred design & evaluation of AI for health equity

### Master of Mathematics (MMath), Computer Science

Sep 2024 - Aug 2026

University of Waterloo

- *Advisor:* Dr. Edith Law
- Thesis-based program in HCI where I designed a virtual patient system for medical education through interviews with stakeholders in standardized patient training program

### B.Sc. Joint Honours in Mathematics and Computer Science, Biochemistry Minor

Sep 2020 - May 2024

Mount Allison University

- *Advisors:* Dr. Matthew Betti, Dr. Laurie Ricker
- Won full-ride scholarship, graduated honours with distinction, Dean's List throughout

## Research Experience

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### Graduate Researcher

Sep 2024 - present

*Advisor:* Dr. Edith Law, *University of Waterloo*

- Developed and ran experiential AI literacy workshops with healthcare professionals across Ontario

### Healthcare AI Independent Research Study

Jan - Apr 2024

*Title:* Large Visual-Language Models for Dermatology Diagnosis

*Advisor:* Dr. Kathleen C. Fraser, *National Research Council Canada*

- Evaluated performance of open- and closed-source visual-language models for diagnosing dermatological skin conditions on varying skin tones via different prompting techniques

### Formal Methods Independent Research Study

Sep 2023 - Apr 2024

*Title:* Attack Detection on ICU IoMT via Formal Methods and Machine Learning

*Advisor:* Dr. Laurie Ricker, *Mount Allison University*

- Developed a model to detect attacks on an Intensive Care Unit (ICU) Internet of Medical Things (IoMT) network via formal methods (dynamic automata) and machine learning

### Undergraduate Summer Research Intern

May - Aug 2023

*Advisor:* Dr. Amber L. Simpson, *Queen's University*

- Developed project on analyzing racial and gender biases in clinical notes, presented this work at the Science Atlantic MSCS Conference, winning first place research award in computer science
- Co-wrote and hosted [The Responsible Use of AI](#) podcast, interviewed 3 multidisciplinary scholars on the implications of AI technologies and tools to ensure it is universally beneficial

### AI4Good Lab Fellow

May - Jun 2022

MILA

- A 7-week AI & ML training program for gender minorities with a focus on applications for social good
- Pitched and developed [heAlthics](#), a research project evaluating racial bias in skin condition and cancer detection AI models by analyzing the diversity and generalizability of datasets and proposing solutions to mitigate the bias

### Summer Research Student

May - Jun 2021; Jul - Aug 2022

*Advisor:* Dr. Matthew Betti, *Mount Allison University*

- Mathematically modeled and analyzed COVID-19 data using a modified SIR model to predict future infection rates in Canada. Model utilized by Public Health Canada
- Modified a mathematical ODE model of honeybee colonies to analyze the threshold for hygienic behaviour for multiple hives

## Publications

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Aakanksha Khandwaha and Edith Law. 2026. *An Experiential Approach to AI Literacy*. In Data Literacy for the 21<sup>st</sup> Century (Workshop Paper, CHI '26).

Aakanksha Khandwaha. 2026. *Are we missing the 'care' in Healthcare AI?* In Towards Relationship Centred-Care with AI (Workshop Paper, CHI '26).

Aakanksha Khandwaha and Edith Law. 2025. *Designing a Simulated Patient System for Medical Education*. In Proceedings of the 2025 ACM Conference on Conversational User Interfaces (CUI '25).

## Academic Presentations

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Aakanksha Khandwaha and Edith Law. 2025. *Designing a Simulated Patient System for Medical Education*. Annual Canadian Celebration of Women in Computing Conference (Lightning Talk, CAN-CWiC '25).

Aakanksha Khandwaha and Amber Simpson. 2023. *Analyzing Racial and Gender Bias in MIMIC-IV Clinical Notes via Sentiment Analysis*. Science Atlantic Mathematics & Statistics and Computer Science Conference (MSCS '23). ★ *best presentation award (1<sup>st</sup> place)*

Aakanksha Khandwaha and Matthew Betti. 2021. *Effectiveness of Non-Pharmaceutical Interventions during COVID-19*. Mount Allison University Student Undergraduate Research Fair (SURF '21).

## Theses and Dissertations

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Aakanksha Khandwaha. 2026. *The Design and Development of a Virtual Patient System for Medical Education*. Master's Thesis. UWSpace.

## Awards and Recognitions

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| 2026        | <b>Provost's Distinguished Inclusivity Doctoral Entrance Award</b> (uWaterloo) — \$5,000<br>Awarded to top female PhD students based on academic excellence                                                                                                |
| 2025        | <b>Queen Elizabeth II Scholarship in Science &amp; Technology (QEII-GSST)</b> (uWaterloo) — \$15,000<br>Awarded to top graduate students provincially based on academics, research potential, and leadership                                               |
| 2025        | <b>President's Graduate Scholarship</b> (uWaterloo) — \$10,000<br>Awarded to top graduate students who hold major federal or provincial awards                                                                                                             |
| 2024        | <b>Math Domestic Graduate Scholarship</b> (uWaterloo) — \$1,000<br>Awarded to incoming Canadian graduate students                                                                                                                                          |
| 2024        | <b>McCall McBain Regional Award</b> (McGill University) — \$5,000<br>Awarded to exceptional students to pursue graduate studies at any public Canadian university                                                                                          |
| 2024        | <b>Scotiabank Graduate Scholarship in Data Science</b> (uWaterloo, <i>declined</i> ) — \$10,000<br>Awarded to two outstanding graduate students in MMath Data Science or MDSAI program                                                                     |
| 2023        | <b>Science Atlantic MSCS Conference Undergraduate Research Award (First Place)</b><br>Awarded for Computer Science research presentation on racial & gender bias in clinical notes                                                                         |
| 2023        | <b>David Gilchrist Chalmers Memorial Prize</b> (Mount Allison University) — \$4,600<br>Awarded to a Math/CS student who has demonstrated high professional potential                                                                                       |
| 2020 - 2024 | <b>Dean's List</b> (Mount Allison University)<br>Recognition given to full-time students who have maintained a first-class average ( $\geq 3.7/4.3$ )                                                                                                      |
| 2020 - 2024 | <b>Marjorie Young Bell Scholarship</b> (Mount Allison University) — \$56,000<br>Awarded to exceptional high school graduates who demonstrate strong academic ability and leadership capability (\$12,000 per year and \$8,000 for personalized internship) |

## Teaching Experience

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### Computer Science Teaching Assistant (TA)

Sep 2024 - present

University of Waterloo

CS497/HLTH480: AI Transformation in Organizations: Healthcare - Fall '26

CS492/692: Social Implications of Computing - Winter '25

CS449/649: Human-Computer Interaction - Fall '24, Spring '25

### Computer Science, Mathematics, and Biochemistry Teaching Assistant (TA)

Sep 2021 - Apr 2024

Mount Allison University

- Organized, prepared, and lead TA training for all computer science TAs during Fall '23

COMP1731: Programming Techniques and Algorithms - Winter '23, Winter '24

Math & Computer Science Help Centre - Fall '22, Winter '23, Winter '24

COMP1631: Intro. to Computer Science - Fall '21, Winter '22, Fall '22, Fall '23

MATH1511: Applied Calculus - Fall '21

BIOC1001: Introduction to Biochemistry - Fall '21

## Mentoring

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Sep 2025 - Aug 2026     Nate MacLeod - NSERC URSRA & Undergraduate Research Fellow  
University of Waterloo

Sep - Dec 2024     Sinry Dong, Rachel Philipose, Saivenkat Lohit Jilla - CRA UR2PhD Mentee  
University of Waterloo

## Academic Service

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### Graphics Interface (GI) 2026 - Student Volunteer Chair

Jun 2025

- Managed and supported SVs with their tasks during the conference
- Worked closely with conference chair to lead SV task development, recruitment and logistics

### ACM Conversational User Interfaces (CUI) 2025 - Student Volunteer

Jul 2025

- Assisted with organizing paper panels and workshops, ran conference registration booth, and arranged banquet

### Computing Research Association (CRA) UR2PhD Program - Graduate Mentor

Sep - Dec 2024

University of Waterloo

- Developed research mentorship skills through online graduate student mentor training course, interacting with other graduate students across North America
- Mentored three University of Waterloo undergraduate students, taught HCI concepts through research project developing LLM-driven patient simulations for medical education

## Community Involvement

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### Women in Computer Science (WiCS) - Graduate Representative

Jun 2026 - present

University of Waterloo

- Organized social events such as monthly coffee breaks, crocheting workshop, and rock climbing

### Math Graduate Student Association (GSA) - Events Coordinator

Oct 2024 - Dec 2025

University of Waterloo

- Organized social events such as monthly coffee breaks, crocheting workshop, and rock climbing

### Technovation Waterloo - Speaker & Judge

Feb - Nov 2025

University of Waterloo

- Presented an introductory lecture about artificial intelligence and AI ethics, to middle school girls
- Judged the Waterloo Regional Pitch Event and annual one-day hackathon, awarding top projects

**Women in Computer Science (WiCS) Conference 2025 - Workshop Speaker**

May 2025

University of Waterloo

- Presented an introductory and interactive workshop on training your own AI for high school and undergraduate students

**MtA Hacks - President**

Sep 2022 - Apr 2024

Mount Allison University

- Lead and organized Mount Allison University's annual computer science conference and hackathon for high school and university students across the Maritimes in 2023 and 2024

**Women in Science Society - Co-President (2022-24), CS Representative**

Sep 2020 - Apr 2024

Mount Allison University

- Developed an elementary school outreach program, hosted a science speaker series, and organized social events for female and gender-diverse science students

**Girl Up Canada - Lead of Socials (External)**

Oct 2020 - Feb 2022

- A youth-led non-profit founded by the United Nations Foundation to advance gender equity and justice through Canadian club chapters.

**Truro Period Collective (prev. PERIOD @ CEC) - Founder & President**

May 2019 - Jun 2020

- Founded the first high school menstrual equity chapter in Atlantic Canada
- Spoke as a panelist at the first Nova Scotia Period Poverty Summit in 2019

**Skills and Certificates**


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| 2024    | <b>Math and Computer Science Teaching Assistant Certificate</b> (Mount Allison University)                                                                                                                                                                                                                                                                                            |
| 2023/24 | <b>Women in Machine Learning (WiML) Career Mentorship Program</b> (Mentee)                                                                                                                                                                                                                                                                                                            |
| 2023    | <b>Bumblekite Machine Learning Summer School in Healthcare and the Biosciences</b><br>Developed machine learning skills for applications in healthcare via lectures and tutorials held at ETH Zurich, Switzerland. Learned from and interacted with senior executives, leaders & top researchers from Emory, ETH Zürich, Cambridge, Roche, Bayer, Biogen, and Novartis, among others. |
| 2023    | <b>Amii Kickstart Program</b>                                                                                                                                                                                                                                                                                                                                                         |
| 2023    | <b>Machine Learning for Health (ML4H) Career Mentorship Program</b> (Mentee)                                                                                                                                                                                                                                                                                                          |